Guo Ye

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Summary

Machine learning researcher building **AI scientists for protein engineering**. I lead DREAM Brain, the AI layer of Northwestern's NSF DREAM Cloud Lab, which designs proteins, runs them through lab automation, and learns from every result. My work spans generative protein design, LLM agents and post-training, GPU infrastructure, and the lab automation that closes the design–build–test–learn loop. My Ph.D. in robot learning gives me years of grounding in **closed-loop autonomous systems**: perception, decision, cloud/edge deployment, monitoring and recovery.

Experience

Data Scientist, Center for Synthetic Biology, Northwestern University Oct 2026 – present

- Building the AI layer of the NSF DREAM Cloud Lab for Protein Engineering, where models design proteins, automation tests them, and every measurement goes back into the models.
- Lead DREAM Brain (below), with the Lucks, Karim and Tullman-Ercek labs as the first science drivers.

Education

2020 – 2026 Ph.D. (Computer Science), **Northwestern University**. Advised by Prof. Han Liu

2018 – 2020 M.S. Computer Science, McCormick School of Engineering, **Northwestern University** (3.78/4.0)

2014 – 2018 B.S. Computer Science, Yingcai Honors School, **University of Electronic Science and Technology of China** (3.87/4.0)

Technical Skills

Protein / molecular design	AF3-style co-folding (Protenix, Boltz, PXDesign, OpenDDE); inverse folding (ProteinMPNN / LigandMPNN); protein language models (ESM, PLMSearch); latent-diffusion generative design; docking, multi-state filtering, hotspot analysis
Machine learning	PyTorch; diffusion models, VAEs, transformers; Reinforcement Learning, imitation learning and active learning; foundation-model fine-tuning, transfer and catastrophic-forgetting control; benchmark design and OOD evaluation
Data engineering	Dataset curation, deduplication and provenance across heterogeneous public corpora; experiment tracking (MLflow, Kubeflow); structure and assay data pipelines
Scientific computing	Multi-GPU training (A100 / H100 / RTX 6000 Ada); AWS, Kubernetes, Docker, Ray, SLURM; CI/CD for model deployment
Agentic AI & interfaces	LLM agent pipelines and post-training (SFT, GRPO); automated analysis-to-report and interactive-visualization generation; Python, C/C++, Java, MATLAB, R
Lab automation	Wet-lab workflows; Opentrons / Hamilton liquid-handling integration; digital twins; dual-arm manipulation, LeRobot, ROS; simulation (IsaacGym, PyBullet, Gazebo)

DREAM Brain: AI Scientist for Protein Engineering

Orchestrator & Scientific Process Data 2026 – present

- Architect of DREAM Brain: three nested loops for wet-lab design–build–test–learn, model learning, and recursive self-improvement of the system itself.
- Built the plain-language-to-wet-lab pipeline: a scientist's goal becomes a quantitative spec, scored designs, a DNA vendor order package, and tracked synthesis, expression and assay runs.
- Designed Scientific Process Data and its compiler, turning meeting audio, slides and instrument files into one record per decision (hypothesis, rejected alternatives, pre-registered prediction, outcome) for training and evaluating scientific agents.

Protein Design & Co-folding Toolkit 2026 – present

- Deploying AF3-class models (Protenix, Boltz, PXDesign, OpenDDE) as interchangeable components behind one design interface.
- Built a de novo design pipeline chaining RFdiffusion, LigandMPNN, ESM, AutoDock and XGBoost, plus active learning over variant libraries for the next design round.
- Fine-tuned a block-level latent-diffusion model for unified small-molecule / peptide / binder design; built a protein-language-model embedding search (PLMSearch) that recovers homologs sequence identity misses.

Literature & Evidence Agent 2026 – present

– Built a provenance-aware evidence agent over the project literature with a verifiable question–answer benchmark, where data checks, training reward and evaluation share one scorer.

– Post-trained open models (Qwen3) with SFT and GRPO; the evidence layer caught conflicting residue-numbering conventions across papers before designs were ordered.

Lab Automation & Digital Twin 2025 – present

– Built a digital twin of the DREAM biofoundry that runs liquid handling, thermocycling, sealing and plate transfers in simulation, with a live command log and step-failure monitoring; every action and reading is logged as model-ready data.

– Designed the DNA preparation workflow on Opentrons as the front end of cell-free expression; automated dexterous steps with a dual-arm AgileX manipulator trained by imitation learning (LeRobot).

Robot Learning & ML Infrastructure

DreamTacVLA: Tactile Vision-Language-Action Models 2025

– Gave a VLA high-resolution tactile input aligned to vision, plus a tactile world model that imagines contact before acting (Think–Dream–Act); up to 95% success on contact-rich manipulation, +22.3% over ablations, trained on 2M tactile frames from a digital twin and real robots.

DOS®: A Deployment Operating System for Autonomous Agents Jan 2022 – Sep 2022

– Built DOS®, a cloud/edge MLOps platform on AWS for reliably operating data-driven autonomous agents in production and simulation: CI/CD for model rollout, versioned storage reservoirs, failure recovery, and packaged executables.

– Adopted by lab members for reproducible, fully automated operation; demonstrated on a swarm of mobile robots. Published at *ICRA 2024*.

Composing Pre-trained Foundation Models for Closed-Loop Control Dec 2023 – Sep 2024

– Fed the outputs of frozen perception foundation models as "prompts" into a decision transformer, reusing pre-trained models across downstream tasks without retraining each one.

– Deployed on a `Unitree GO1` across meadow, mud and ladder terrain, demonstrating simulation-to-reality transfer of the learned policy.

Publications

Guo Ye, Zexi Zhang, Xu Zhao, Shang Wu, Haoran Lu, Shihan Lu, and Han Liu (2025). "Learning to Feel the Future: DreamTacVLA for Contact-Rich Manipulation". In: *arXiv preprint arXiv:2512.23864*.

Guo Ye, Muye Jia, Mingfu Liang, Matthew Elwin, and Han Liu (2024). "Visual Prompt via Pre-trained Model for Quadrupedal Locomotion". In: Preprint.

Guo Ye, Qinjie Lin, Zening Luo, and Han Liu (2024). "DOS®: A Deployment Operating System for Robots". In: 2024 IEEE International Conference on Robotics and Automation (ICRA).

Lin, Qinjie, Guo Ye, and Han Liu (2023). "EMS®: A Massive Computational Experiment Management System towards Data-driven Robotics". In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE.

Luo, Haozheng, Ruiyang Qin, Chenwei Xu, Guo Ye, and Zening Luo (2023). "Open-Ended Multi-Modal Relational Reasoning for Video Question Answering". In: *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, pp. 363–369.

Guo Ye, Qinjie Lin, Wanxin Jin, Zhaoran Wang, Haozheng Luo, Tim Tsz-Kit Lau, Cheng Zhou, and Han Liu (2022). "RobLAX: A Differentiable Robotics Framework for Physics Augmented Reinforcement Learning". In: Preprint. URL: https://sites.google.com/u.northwestern.edu/roblax/home.

Guo Ye, Han Liu, and Biswa Sengupta (2022). "Learning to Infer Belief Embedded Communication". In: *arXiv preprint arXiv:2203.07832*.

Lin, Qinjie, Guo Ye, Jiayi Wang, and Han Liu (2022). "RoboFlow: a Data-centric Workflow Management System for Developing AI-enhanced Robots". In: *Conference on Robot Learning (CoRL), Blue Sky oral*. PMLR, pp. 1789–1794.

Guo Ye, Qinjie Lin, Tzung-Han Juang, and Han Liu (2020). "Collision-free Navigation of Human-centered Robots via Markov Games". In: *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, pp. 11338–11344.

Chen, Binghong, Bo Dai, Qinjie Lin, Guo Ye, Han Liu, and Le Song (2020). "Learning to plan in high dimensions via neural exploration-exploitation trees". In: *2020 International Conference on Learning Representations (ICLR)*. Spotlight.

He, Shiyuan, Guo Ye, Mengqiu Hu, Yang Yang, Fumin Shen, Heng Tao Shen, and Xuelong Li (2018). "Learning binary codes with local and inner data structure". In: *Neurocomputing* 282, pp. 32–41.